Pranav Sanjeev Jain

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Education

University of Michigan — Ann Arbor, MI, USA
Masters of Science in Robotics, ***Cumulative GPA 4.0 (May 2026)*** — *Aug 2025 – Dec 2026*
- **Relevant Coursework:** Robotic Systems Lab, Self-Driving Cars, Multi-Robot Systems, Mobile Robotics

Worcester Polytechnic Institute — Worcester, MA, USA
Bachelor of Science in Robotics Engineering, ***GPA 4.0 (High Distinction)*** — *Aug 2022 – May 2025*
- **Dean's List** (All Semesters) & **Charles O. Thompson Scholar** (March 2023)
- **WPI Presidential Scholarship** (Aug 2022 - May 2025)
- **Relevant Coursework:** Mechanical Applications in Robotics, Sensing and Perception in Robotics, Robot Manipulation, Robot Navigation, Soft Robotics, Metal Additive Manufacturing, Control Engineering, Embedded Computing in Engineering Design, Software Engineering

Technical Skills

Languages: Python, C++, Java, MATLAB, SQL, JavaScript, HTML/CSS
CAD Softwares: Fusion 360, SolidWorks, Autodesk 3ds Max, Altair Inspire, Zbrush, Adobe Substance 3D Painter
Fabrication & Machining: Laser Cutting, Milling, Lathe, 3D printing, Oscilloscope, Soldering
Others: ROS, Gazebo, Rviz, PCB design (KiCad), Git

Experience

Mechanical Engineering Intern, Keolis, Mitsubishi Heavy Industries, Dubai, UAE — May - Aug 2026
- Analyzed Alstom/Kinki Sharyo rolling stock subsystems including propulsion, braking, and bogie systems.
- Shadowed cross-functional engineering teams on diagnostic, preventative & corrective maintenance procedures, learning the four-tier maintenance hierarchy and MEPS depot operations to international safety standards.

Projects

Quadruped Robotic Dogs | *SolidWorks, Fusion 360, 3D Printing, Machining* — Jan - Dec 2026
- Developed a multi-robot quadruped system by designing and prototyping robotic dog components using additive manufacturing techniques and laser cutting.
- Redesigned components using CNC-millable parts, improving durability, scalability & production readiness.

2D Swarm Image Reconstruction | *MATLAB, Multi-Agent Systems, HOCBF-CLF* — Jan - April 2026
- Implemented motion planning, task allocation, & collision-avoidance strategies under unicycle model constraints to convert images into waypoint distributions for multiple robots to recreate visual patterns collaboratively.

Weather-Robust BEV Perception | *Python, CARLA, BEV, U-Net, Self-Driving Vehicles* — Aug - Dec 2025
- Developed a multi-camera Bird's Eye View (BEV) occupancy estimation, using U-Net-based architecture, under diverse weather conditions including clear, rainy, & foggy environments to improve robustness of self-driving perception systems.

5-DOF Arm: RGB-D Vision Pick-and-Place | *Python, ROS, DH parameters* — Aug - Dec 2025
- Developed a ROS-based arm manipulation system that used forward & inverse kinematics to sort blocks using an external camera for block position, height & color detection.
- Utilized Denavit-Hartenberg (DH) parameters to calculate transformation matrices for 5 DOF arm.

Soft Robotic Eel | *Arduino, ESP32, 3D Printing* — Aug 2024 - May 2025
- Designed a modular eel for aquatic locomotion, and worked on depth control and sensor integration.
- Modeled segment curvature to generate sinusoidal body-wave locomotion for anguilliform swimming.

TurtleBot: LiDAR SLAM & Maze Navigationn | *Python, GitHub, ROS, SLAM, AMCL* — Oct - Dec 2024
- Developed a ROS-based autonomous navigation system to map, localize, & navigate through a maze using GMapping, SLAM, Adaptive Monte Carlo Localization, & A* path planning algorithms.

Others

Honor Society: Tau Beta Pi — Apr 2024 - Present
Guest speaker at WPI on AI & ChatGPT — Apr 2024
- Discussed AI advancements, challenges, and its impacts at home and workplace.